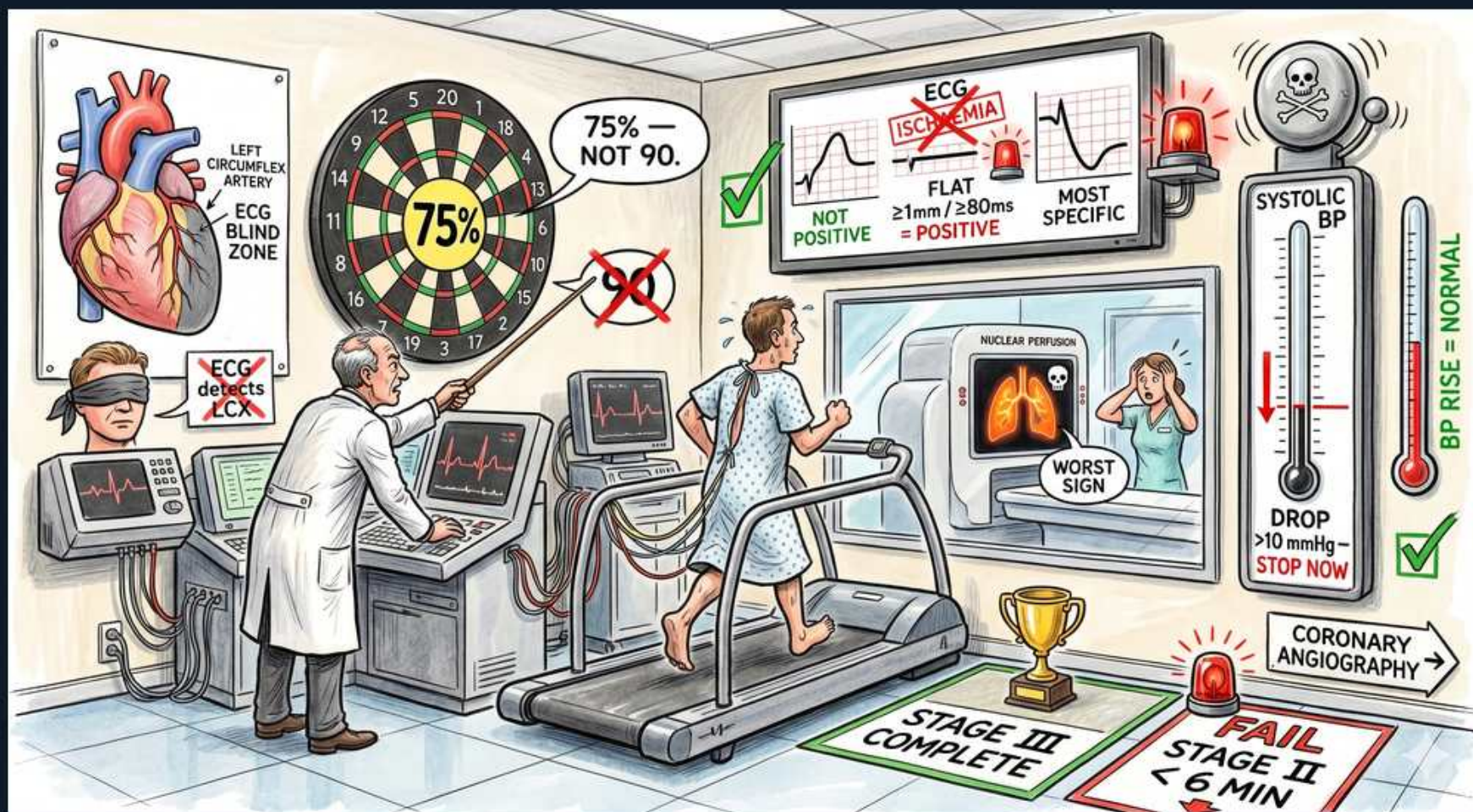


# Stress Testing — Interpretation, Termination & Risk



## 1. Left circumflex blind zone

### WHAT YOU SEE

Heart poster: 'LEFT CIRCUMFLEX ARTERY — ECG BLIND ZONE'

### WHAT IT ENCODES

The LCx territory (lateral/posterior wall) is the classic ECG blind spot — ischemia there can be electrically silent on a standard 12-lead. A normal exercise ECG does not exclude LCx disease; posterior leads or imaging are needed.

### BOARD TRAP

Calling a normal stress ECG 'no ischemia' — a true posterior/LCx event may show only reciprocal anterior ST depression or nothing at all.

## 2. Blindfolded ECG tech

### WHAT YOU SEE

Blindfolded man at console: 'ECG detects ↓↓'

### WHAT IT ENCODES

Exercise ECG has limited sensitivity (~68%) and specificity (~77%). It detects only marked ischemia. Imaging stress (echo/nuclear) is preferred when baseline ECG is uninterpretable (LBBB, paced, LVH with strain, digoxin, WPW).

### BOARD TRAP

Ordering exercise ECG in a patient with baseline LBBB or paced rhythm — use vasodilator nuclear or dobutamine echo instead.

## 3. 75% not 90% pre-test

### WHAT YOU SEE

Dartboard '75%' with speech bubble '75% NOT 90'

### WHAT IT ENCODES

Stress testing is most useful at INTERMEDIATE pre-test probability. Bayes: a positive test in a low-probability patient is mostly false-positive; in very-high-probability patients (~90%) go straight to angiography — the test rarely changes management.

### BOARD TRAP

Stress-testing a patient with typical angina and very high pre-test probability instead of proceeding to coronary angiography.

## 4. ≥1mm flat/downslope ST

### WHAT YOU SEE

Monitor: 'NOT POSITIVE ... FLAT ≥1mm / ≥80ms = POSITIVE'

### WHAT IT ENCODES

A positive exercise ECG = ≥1 mm horizontal or downsloping ST depression 80 ms after the J-point. Upsloping ST depression is NOT diagnostic. ST elevation in non-Q leads suggests transmural ischemia/spasm.

### BOARD TRAP

Counting upsloping ST depression as a positive test — only flat or downsloping ≥1 mm qualifies.

## 5. Imaging = most specific

### WHAT YOU SEE

Banner 'MOST SPECIFIC'

### WHAT IT ENCODES

Stress imaging (nuclear MPI or stress echo) adds specificity and localizes ischemia. Reversible perfusion defect = ischemia; fixed defect = scar/infarct. Echo wall-motion abnormality with stress = ischemia.

### BOARD TRAP

A FIXED defect is scar, not active ischemia — it does not warrant revascularization for symptom relief the way a reversible defect does.

## 6. Nuclear perfusion — worst sign

### WHAT YOU SEE

Nuclear lungs image labeled 'WORST SIGN'

### WHAT IT ENCODES

Increased lung uptake of thallium (or transient ischemic dilation, large multi-territory reversible defect, drop in EF) signals severe, often left-main/3-vessel disease — high-risk markers that mandate angiography.

### BOARD TRAP

Treating a large reversible anterior defect or TID as routine — these are high-risk findings requiring prompt invasive evaluation.

## 7. Systolic BP drop ≥10 — STOP

### WHAT YOU SEE

Thermometer 'SYSTOLIC BP ↓ DROP >10 mmHg STOP NOW'

### WHAT IT ENCODES

A FALL in systolic BP ≥10 mmHg during exercise (exertional hypotension) signals severe ischemia/LV dysfunction — an absolute reason to terminate. Normal response is a progressive rise in systolic BP.

### BOARD TRAP

Continuing the test despite a falling systolic BP — exertional hypotension is a marker of left-main/severe multivessel disease.

## 8. BP rise = normal

### WHAT YOU SEE

Thermometer 'BP RISE = NORMAL' with green check

### WHAT IT ENCODES

Normal hemodynamic response is a steady rise in systolic BP and heart rate with exercise. Failure to augment HR (chronotropic incompetence) or BP is abnormal and prognostically adverse.

### BOARD TRAP

Mistaking a blunted/flat HR response on a beta-blocked patient as ischemia — but true chronotropic incompetence off rate-limiting drugs is high-risk.

## 9. Coronary angiography next

### WHAT YOU SEE

Arrow 'CORONARY ANGIOGRAPHY'

### WHAT IT ENCODES

High-risk stress findings (exertional hypotension, early/marked ST depression, large reversible defect, EF drop, ↑lung uptake) → proceed to coronary angiography ± revascularization.

### BOARD TRAP

Sending a high-risk positive test home on medical therapy alone instead of to the cath lab.

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## 10. Stage III complete

### WHAT YOU SEE

Green box 'STAGE III COMPLETE' + trophy

### WHAT IT ENCODES

Good exercise capacity (completing later Bruce stages, high METs, Duke Treadmill Score) is strongly protective — exercise tolerance is one of the best prognostic predictors regardless of imaging.

### BOARD TRAP

Ignoring excellent functional capacity — high METs achieved predicts good outcome even with minor ST changes.

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## 11. Fail Stage II <6 min

### WHAT YOU SEE

Red box 'FAIL STAGE II <6 MIN' with siren

### WHAT IT ENCODES

Inability to complete early stages / low METs / ischemia at low workload (Duke Treadmill Score  $\leq -11$  = high risk) predicts severe disease and poor prognosis — refer for angiography.

### BOARD TRAP

Reassuring a patient who developed angina and ST depression at a low workload — early ischemia is a high-risk finding.

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## 12. ECG ISCHEMIA banner

### WHAT YOU SEE

Top monitor labeled 'ECG ISCHEMIA' with siren/skull

### WHAT IT ENCODES

Absolute termination indications: ST elevation >1 mm in non-Q leads, drop in SBP with signs of ischemia, sustained VT, signs of poor perfusion (cyanosis/pallor), patient's request, or technical failure to monitor.

### BOARD TRAP

Pushing through sustained VT or ST elevation — these are absolute stop indications, not relative.

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